

# ALI NIKKHAH

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## SUMMARY

Applied ML researcher & engineer (Sharif B.Sc. Electrical Engineering 2020-2025, GPA 3.65; M.Sc. AI from Sep 2025) specialising in agentic RAG, multimodal vision-language, self-hosted LLM serving, and GPU runtime optimisation. Shipped LLM + CV pipelines at marketplace scale, self-hosted Graph RAG assistant for a national credit bureau, and BNPL risk platforms. Research focus: LLM reasoning and hallucination mitigation. — Variant: Hybrid AI

## TECHNICAL SKILLS

**Languages:** Python, C++, Go, SQL, R, Bash, TypeScript  
**ML/Vision:** PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper  
**Agentic:** LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE  
**LLM Serving/GPU:** vLLM, TGI, AWQ, GPTQ, KV-cache  
**Data/MLOps:** Spark, Airflow, Kafka, Docker, Kubernetes, Ray

## PROFESSIONAL EXPERIENCE

### Iran Credit Scoring Bureau

Tehran, Iran

*ML Engineer — Self-Hosted Graph RAG Assistant (Concurrent)*

*Apr 2026 == Present*

- Architected self-hosted Graph RAG assistant: Neo4j knowledge graph + vector retrieval fused into an agent that answers analyst queries over credit-scoring domain data with grounded citations.
- Ran self-hosted open-weight LLMs (vLLM/TGI) on GPU; profiled and optimised GPU runtime — KV-cache, batching, quantization (AWQ/GPTQ) — cutting per-token latency and GPU memory footprint.
- Implemented Neo4j tool-calls (Cypher generation + schema-aware retrieval) so the assistant executes graph traversals instead of hallucinating structured facts.
- Added hallucination guardrails — retrieval grounding, citation enforcement, deterministic fallback — for a regulated financial environment.
- *Stack:* Python, Neo4j, Cypher, Graph RAG, LangGraph, LangChain, FAISS, vLLM
- *Most recent Iran position; overlaps Sharif M.Sc. AI (Sep 2025) — full-time industry alongside full-time graduate study*

### Turquoise Digital

Tehran, Iran

*Senior Data Scientist & Data Engineer (Concurrent)*

*Sep 2025 == Mar 2026*

- Centralized lake from raw CDC via PeerDB + Trino/Parquet on 5TB/day, <60s freshness; Airflow + PySpark DAGs cut memory 60%, governed by Great Expectations + Prometheus/Grafana.
- Built default-risk ensemble (XGBoost + time-series Transformers) — AUC 0.87, -22% defaults; 50+ concurrent mSPRT A/B tests with canary & auto-rollback, presented to C-suite.
- Shipped Whisper/wav2vec2.0 + LangGraph agentic chatbot — -78% ticket volume, 94% intent accuracy; deterministic fallback + policy guardrails for auditability.
- Drove portfolio allocation & risk engine (Sharpe +0.42) with streaming ClickHouse signals; LlamaIndex retrieval graph for ranking features.
- *Stack:* Python, PyTorch, Spark, Trino, PeerDB, Airflow, Great Expectations, Prometheus
- *Concurrent with Advanced Analytics Australia (Remote Contract) — 7-month overlap, split focus: Turquoise BNPL/risk/whisper vs Advanced edge CV*

### Advanced Analytics Australia

Remote · Australia (Tehran-based)

*Computer Vision / ML Platform Engineer (Edge) (Concurrent)*

*Sep 2025 == Apr 2026*

- Deployed YOLO + ViT quantized (TensorRT/TF Lite) on Triton — 99.2% uptime <50ms p99, 3.5x speedup, exactly-once Kafka, 30 FPS over 100+ streams.
- Built edge MLOps with Triton model repository, zero-downtime rolling updates, automated rollback, Prometheus/Grafana per-site health.
- Cut false positives -35% via attention-guided NMS + SNR monitoring; broker architecture for 12 distributed sites.
- *Stack:* Python, OpenCV, YOLO, ViT, TensorRT, TFLite, Triton, Kafka
- *Parallel remote contract alongside Turquoise — edge CV track*

### Sharif University — Robust & Interpretable ML Lab

Tehran, Iran

*Research Collaborator — Explainable & Robust AI (Concurrent)*

*Jul 2025 == Sep 2025*

- Benchmarked adversarial + natural shift suites; notebooks surfacing Captum/SHAP attributions, saliency, counterfactuals.
- Documented deploy best-practices for regulated industries with faculty.
- *Stack:* PyTorch, MLflow, Captum, SHAP

– Summer research sprint concurrent with industry roles

### Trinity College Dublin — EmoDub Project

Dublin, Ireland

Machine Learning Researcher — Emotion-Aware Speech Translation (Concurrent)

May 2025 == Nov 2025

- Trained SER front-end (transformer + wav2vec2.0 + Whisper) for affect logits; cross-modal fusion transformer for emotion-consistent decoding.
- Curated emotion-annotated corpora (EmotionNet/EmoDB) with balanced cultural representation; evaluated BLEU, Emotion F1, MOS.
- Collab. with Prof. Siobhán Clarke; reproducible codebase  
<https://github.com/AliNikkhah2001/emodub-emotion-translation>.
- *Stack*: PyTorch, TensorFlow, Hugging Face, librosa, wav2vec2.0, Whisper, EmotionNet, EmoDB
- Part-time remote research concurrent with Digikala + Turquoise + UBC

### Digikala

Tehran, Iran

AI Engineering R&D Lead (Concurrent)

Jan 2025 == Aug 2025

- Led team of 5 defining R&D roadmap + SLOs; shipped MCP with LangGraph cyclic graphs, zero-downtime hot-swapping, health-checked agent composition with deterministic fallback.
- Hybrid retrieval FAISS + Elasticsearch RRF + HyDE + KAG — +23% recall@10, -41% hallucination; multi-tier Redis caching, p99 <200ms.
- LLM serving on vLLM (PagedAttention) + Triton — 3.2x throughput, 60% cost reduction; 200 pods, MIG/HPA/VPA, Prometheus KV-cache dashboards.
- Partnered with infra/product establishing eval pipelines, A/B guardrails, canary rollouts.
- *Stack*: Python, LangGraph, LangChain, LlamaIndex, FAISS, Elasticsearch, ChromaDB, HyDE
- Full-time lead; UBC/Trinity/Sharif were part-time remote (<10h/wk)

### Digikala

Tehran, Iran

AI Engineer (Concurrent)

Nov 2024 == Jan 2025

- Implemented LangGraph orchestration layers with health monitoring; hybrid retrieval with RRF + deterministic fallbacks for auditability.
- Embedded eval telemetry, memory-efficient indexing, response caching; load-tested to 5M queries.
- *Stack*: LangChain, LlamaIndex, FAISS, Elasticsearch, FastAPI
- Promotion track — converted to R&D Lead Jan 2025

### University of British Columbia

Remote · Vancouver, Canada

Research Assistant — Vision-Language Medical Imaging (Ultrasound Report Generation) (Concurrent)

Feb 2025

- Designed ViT/BLIP encoders + T5 generator with contrastive/self-supervised alignment; curated DICOM, de-identified QA corpora.
- Deployed attention/saliency overlays for interpretability; benchmarked BLEU/ROUGE-L + clinical efficacy with radiologists; SOTA on in-house split.
- Preprint in prep: Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning (2025).
- *Stack*: PyTorch, ViT, BLIP, T5, LLaVA, CLIP, OpenCV, MIMIC-CXR
- Long-running part-time research; overlapped industry roles but capped at 10h/wk — primary commitment was Digikala/Turquoise

### L3S Research Center — Leibniz University Hannover

Hannover, Germany

Research Assistant — Video Motion Classification (Concurrent)

Apr 2023 == Feb 2024

- Engineered optical-flow-centric datasets with temporal sampling/augmentation; improved domain generalization across lighting/texture/camera shift.
- Benchmarked 3D CNN + transformer hybrids vs baselines; analyzed attention maps; reduced overfitting, SOTA accuracy.
- *Stack*: PyTorch Lightning, OpenCV, PyAV, TensorBoard, scikit-learn, Multimodal Models
- Part-time RA during B.Sc. final year — no conflict

### HomaCloud

Tehran, Iran

MLOps Engineer (Founding) (Concurrent)

Aug 2021 == Feb 2022

- Provisioned multi-node K8s via Terraform; Ray + MLflow training pipelines, TF Serving + FastAPI — <10 min releases, 99.9% deploy success, 15+ models.
- GitOps with GitHub Actions, canary, spot-instance optimization; Prometheus/Grafana + ELK observability; registry + lineage.
- *Stack*: Kubernetes, Docker, Ray, MLflow, Terraform, ArgoCD, FastAPI, Prometheus
- During B.Sc. — clean sequential after Mapna internship

## SELECTED PROJECTS & RESEARCH

Trinity College Dublin · EmoDub

May 2025 – Nov 2025

Emotion-aware voice translation

– SER front-end + wav2vec2.0 bullets (see experiences)

**University of British Columbia**

Feb 2024 – Nov 2025

*Ultrasound report generation*

**L3S Research Center**

Apr 2023 – Feb 2024

*Texture-free motion intelligence*

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning — *arXiv preprint*  
— *in preparation*, 2025

## EDUCATION

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**Sharif University of Technology**

Tehran, Iran

*M.Sc. Artificial Intelligence    Computer Engineering (full-time, ongoing)*

*Sept 2025 – Present*

**Sharif University of Technology**

Tehran, Iran

*B.Sc. Electrical Engineering    GPA: 3.65*

*Sept 2020 – Sept 2025*